

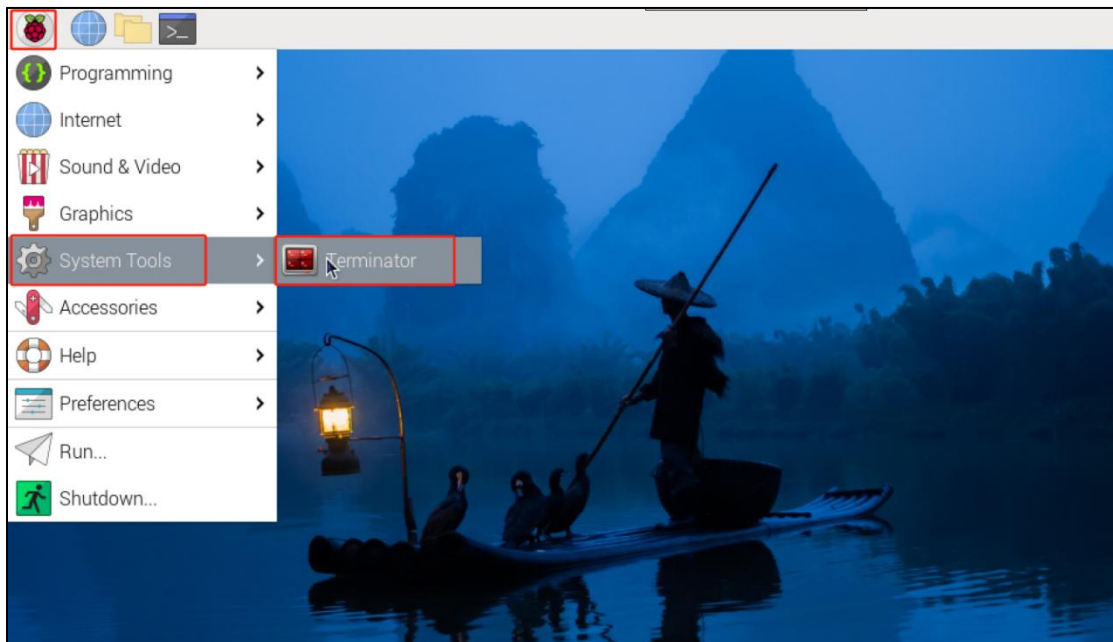
Lesson 20 ROS2 RQT Tool Usage Instruction

1. RQT Instruction

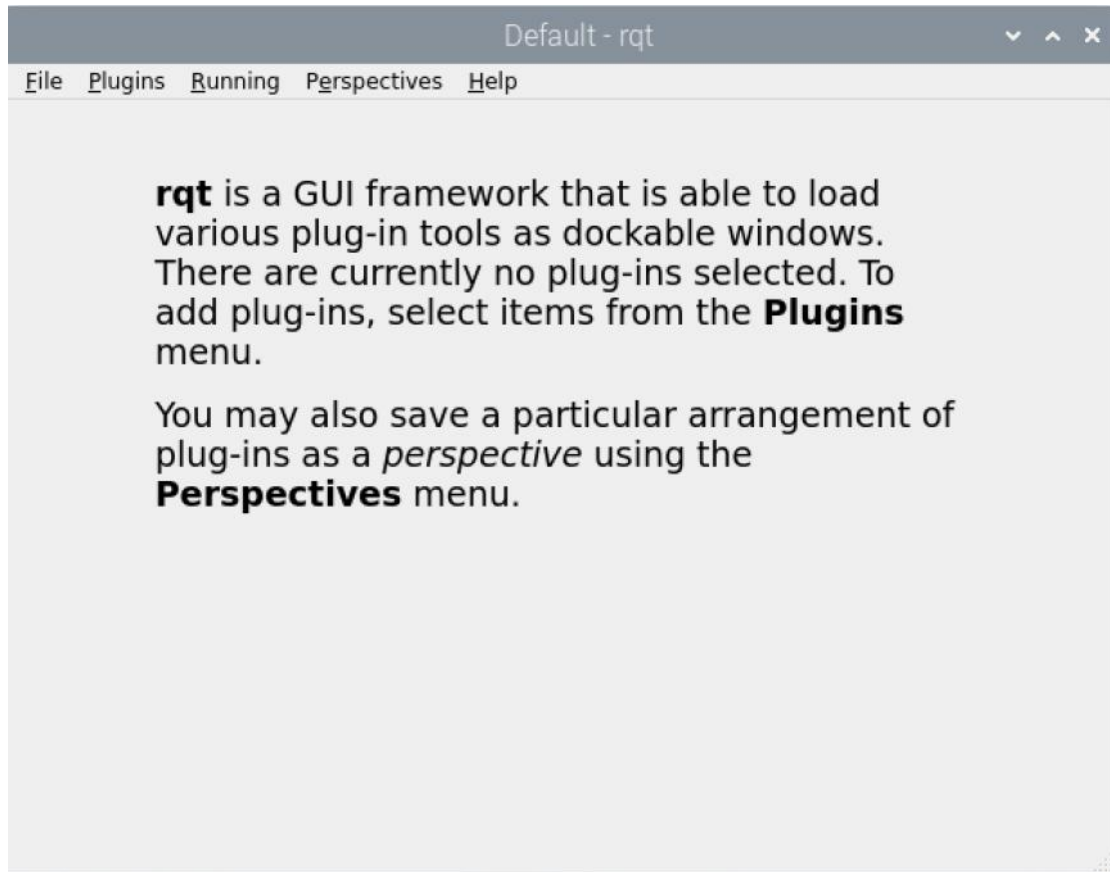
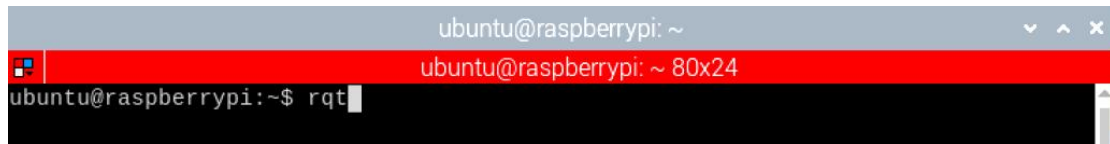
The functionality of Rviz in ROS is already quite powerful. However, in some scenarios, we may prefer simpler, modular visualization tools. For example, displaying images from just one camera. If using Rviz for this purpose, it might feel a bit cumbersome. In such cases, we can turn to another modular visualization tool provided by ROS, called RQT.

Just like Rviz, the name RQT indicates that it is also based on the QT visualization tool.

- 1) Click on  at the top left corner, and then select “**System Tools**→**Terminator**” in sequence.



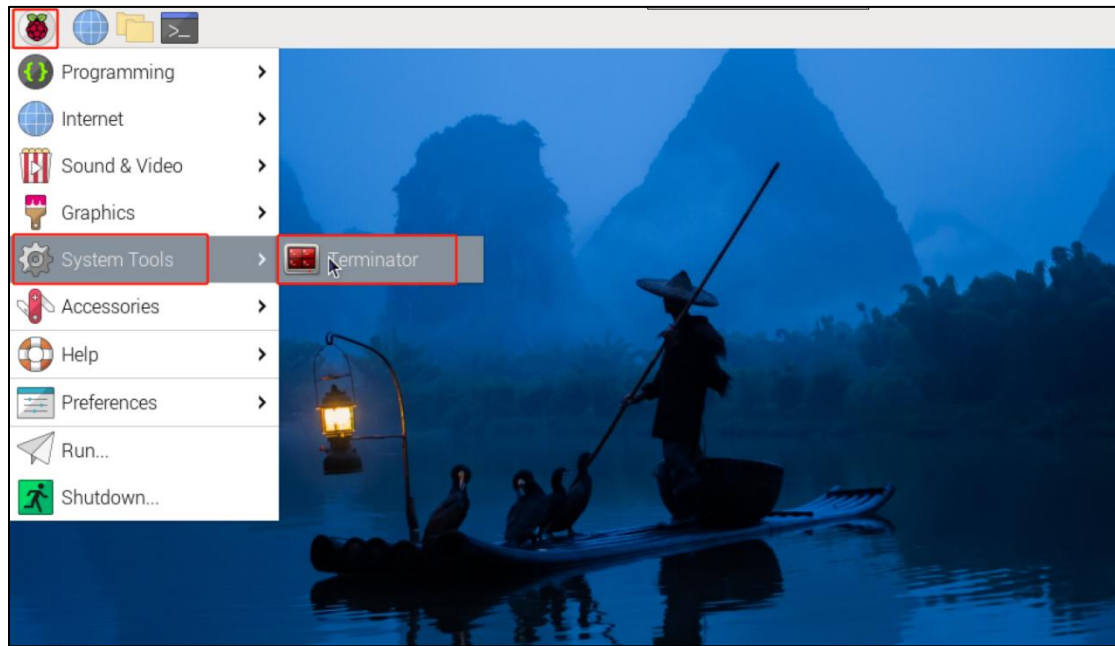
- 2) Enter the command “**rqt**” to open RQT tool.



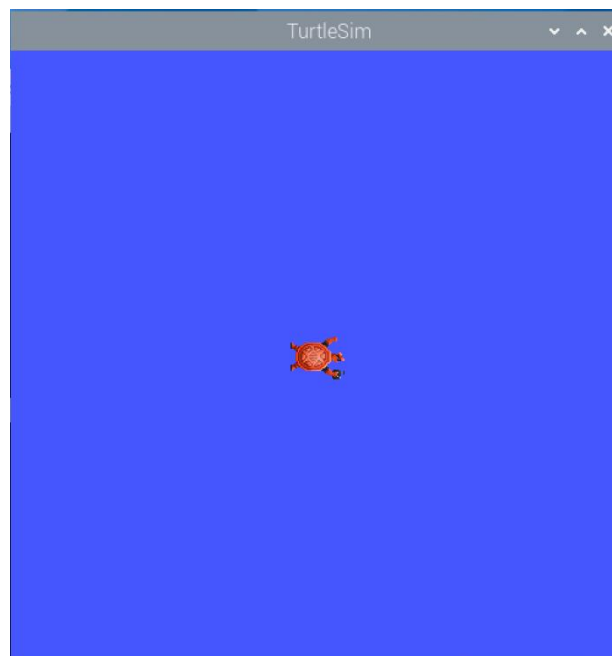
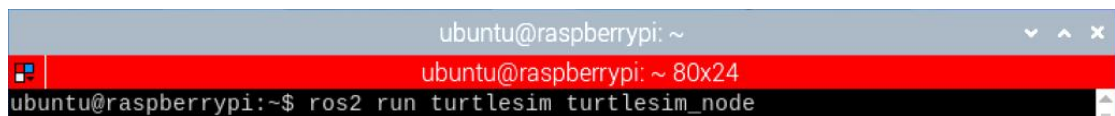
This interface loads many small modules, each capable of implementing a specific small functionality. The commonly used modules contain topics, services, actions, parameters, logs, etc. You can use them according to your requirements, facilitating the ROS2 program debug.

2. Logging Display

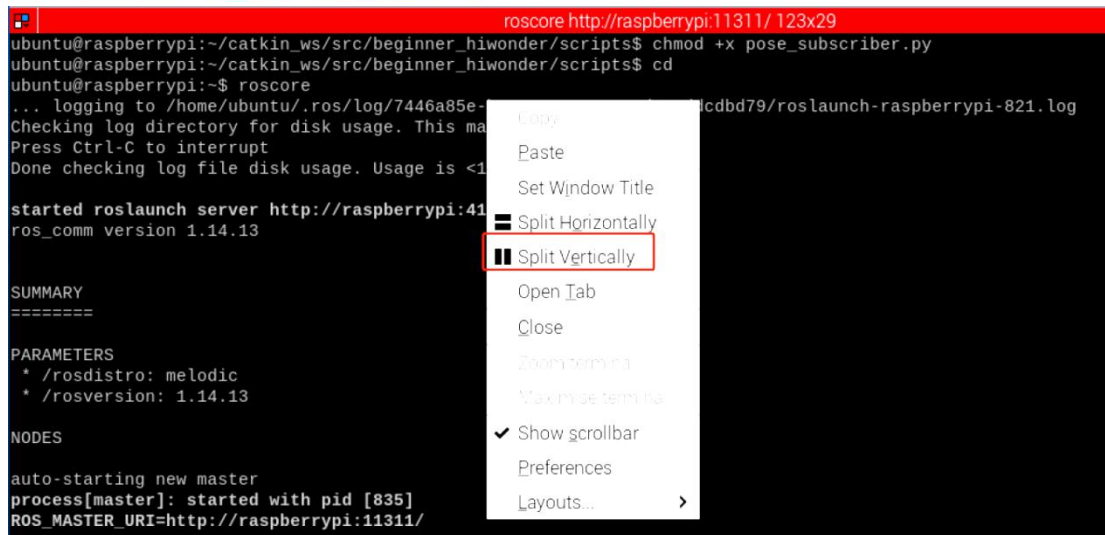
- 1) Click on  at the top left corner, and then select “**System Tools**→**Terminator**” in sequence.



2) Enter the command “**ros2 run turtlesim turtlesim_node**” to launch the TurtleSim node.



3) Right click on a blank space to select “**Split Vertically**” to create a new terminal window.



```

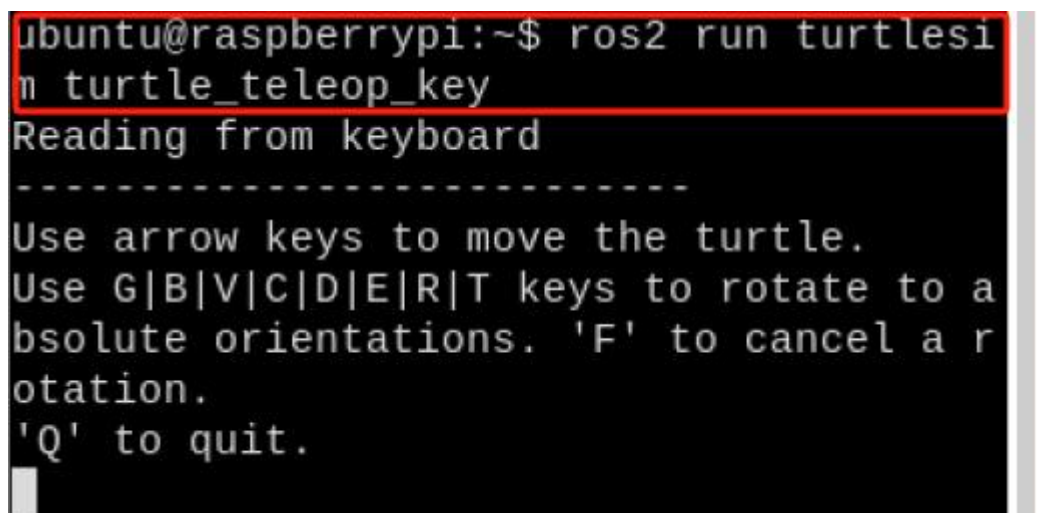
roscore http://raspberrypi:11311/ 123x29
ubuntu@raspberrypi:~/catkin_ws/src/beginner_hiwonder/scripts$ chmod +x pose_subscriber.py
ubuntu@raspberrypi:~/catkin_ws/src/beginner_hiwonder/scripts$ cd
ubuntu@raspberrypi:~$ roscore
... logging to /home/ubuntu/.ros/log/7446a85e-.../roslaunch-raspberrypi-821.log
Checking log directory for disk usage. This may take a while.
Press Ctrl-C to interrupt
Done checking log file disk usage. Usage is <1%

started roslaunch server http://raspberrypi:4141/
ros_comm version 1.14.13

SUMMARY
=====
PARAMETERS
 * /rostdistro: melodic
 * /rosversion: 1.14.13

NODES
auto-starting new master
process[master]: started with pid [835]
ROS_MASTER_URI=http://raspberrypi:11311/
    
```

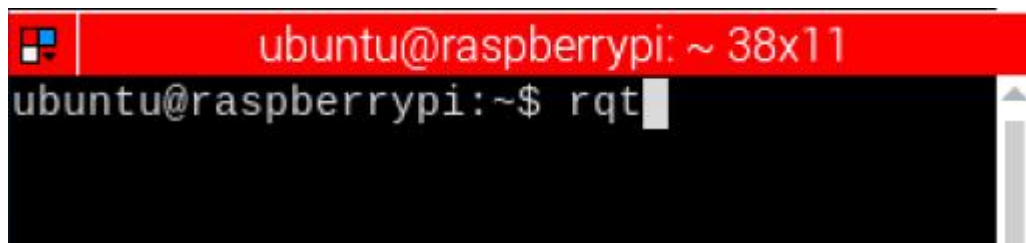
4) Enter the command “**ros2 run turtlesim turtle_teleop_key**” and press Enter to launch the TurtleSim control node.



```

ubuntu@raspberrypi:~$ ros2 run turtlesim turtle_teleop_key
Reading from keyboard
-----
Use arrow keys to move the turtle.
Use G|B|V|C|D|E|R|T keys to rotate to absolute orientations. 'F' to cancel a rotation.
'Q' to quit.
    
```

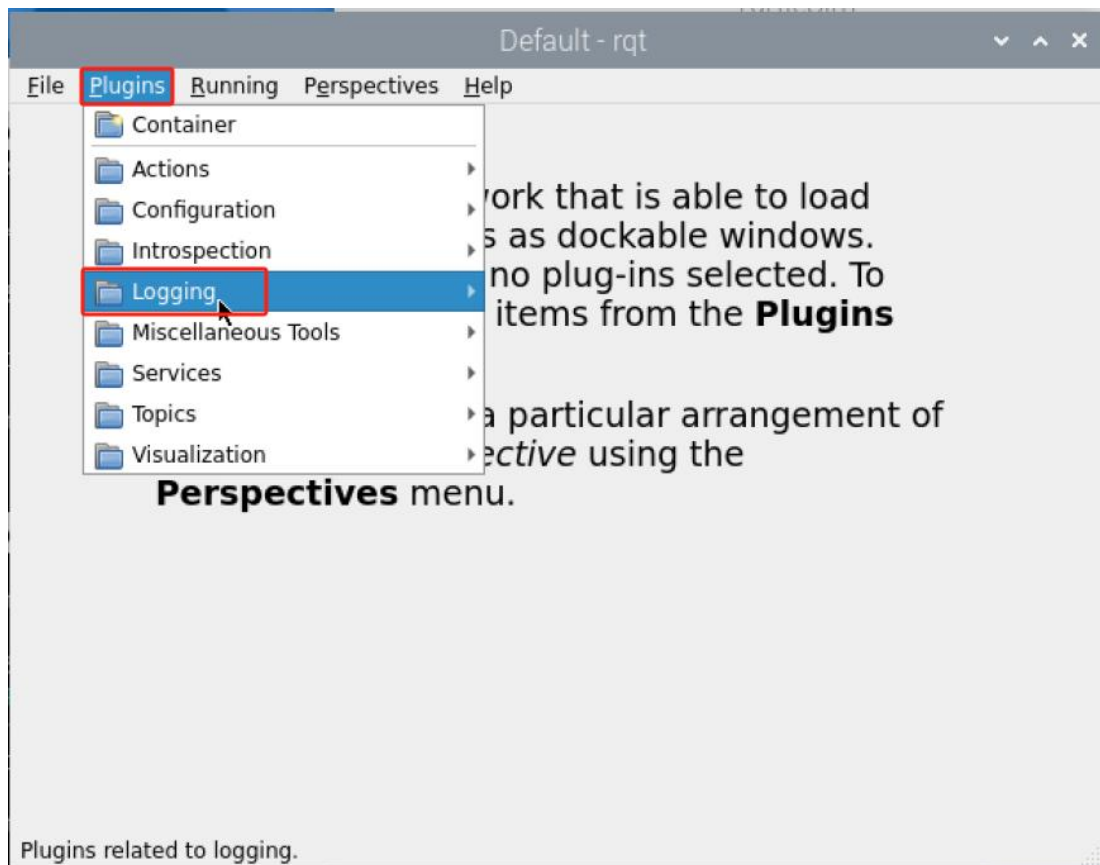
5) Refer to step 3) to create a new window, and then enter command “**rqt**” to open RQT tool.



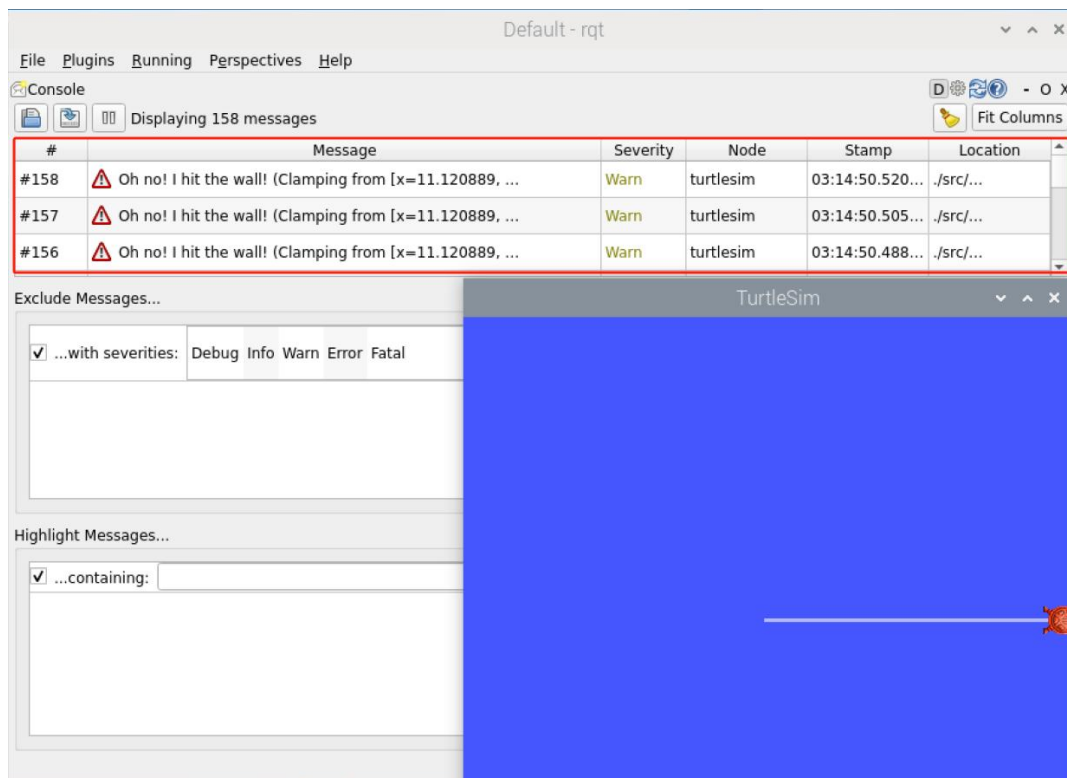
```

ubuntu@raspberrypi: ~ 38x11
ubuntu@raspberrypi:~$ rqt
    
```

6) Select “**Plugins→Logging→Console**” in sequence to open logging display.

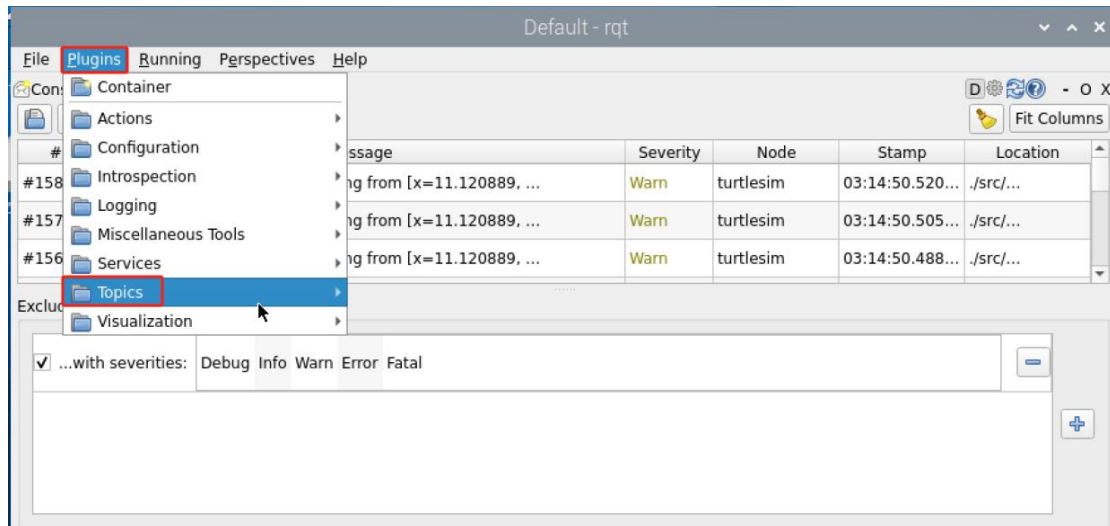


7) Use the "↑ ↓ ←→" keys to control the movement of the turtle. If the turtle touches the wall, log messages will be displayed.

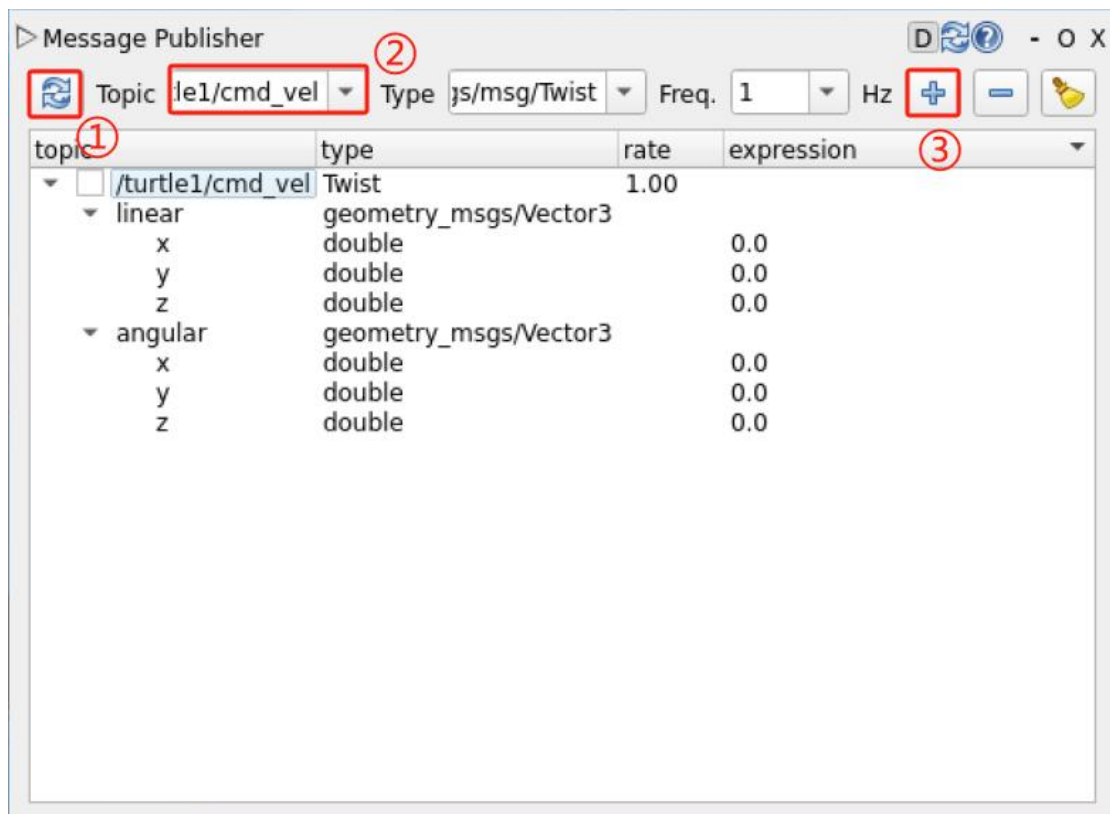


3. Topic Pugin

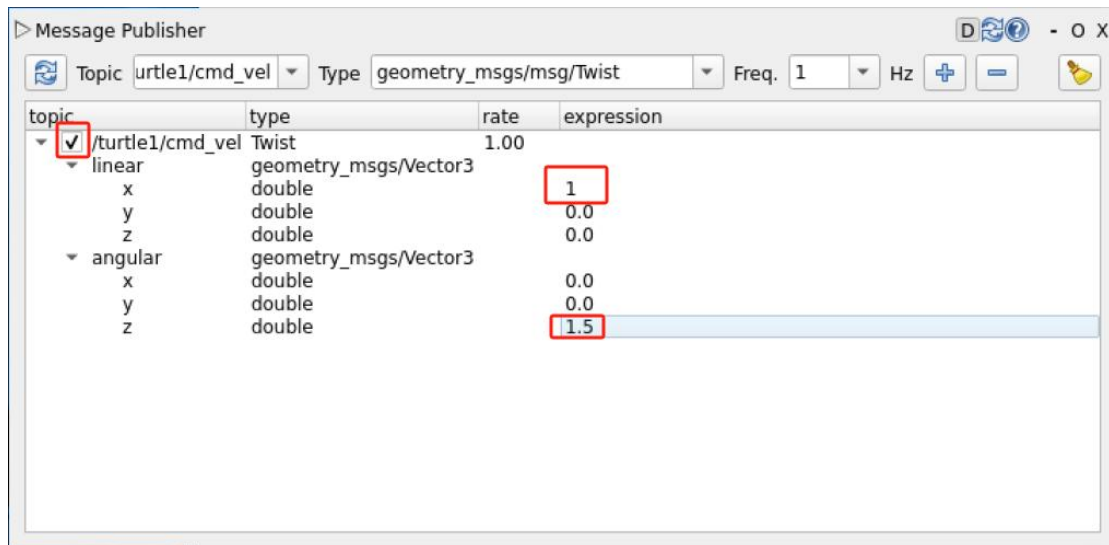
1) Select “**Plugins→Topic→Message Publisher**” in sequence to open the topic plugin.



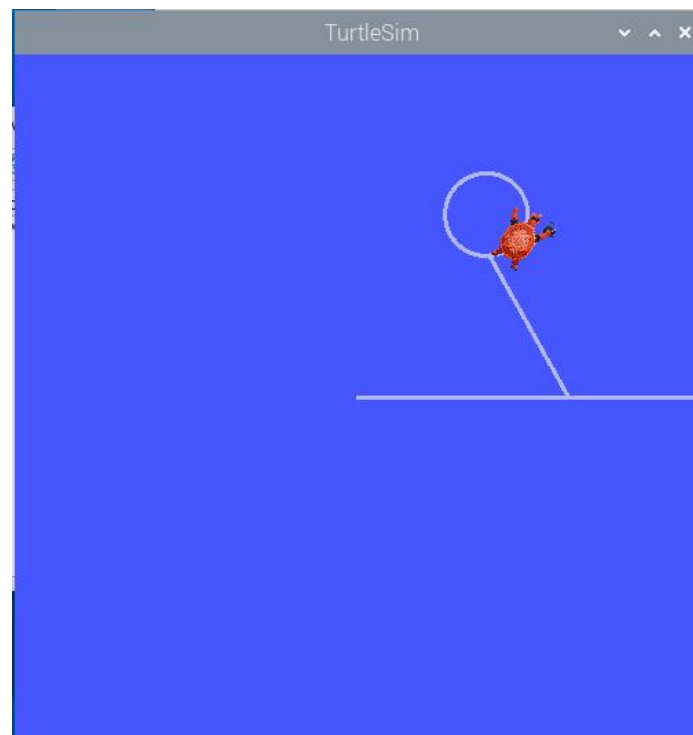
2) Click on , then select the topic of “**/turtle1/cmd_vel**”, then click on .



3) Modify the the linear velocity value X and the angular velocity Z, then select and check.

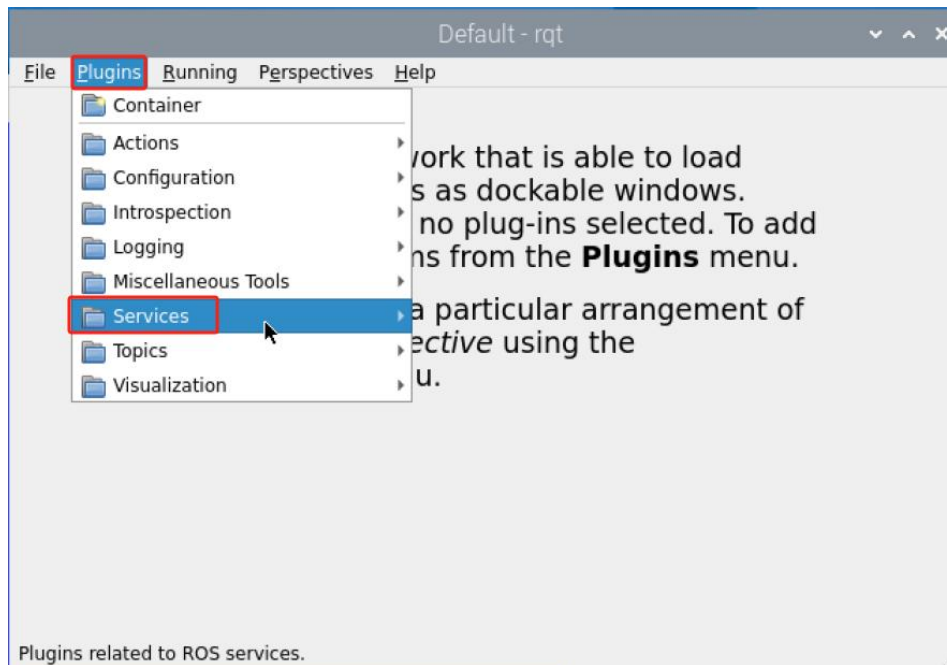


The turtle will move according to the specific linear and angular velocity data.

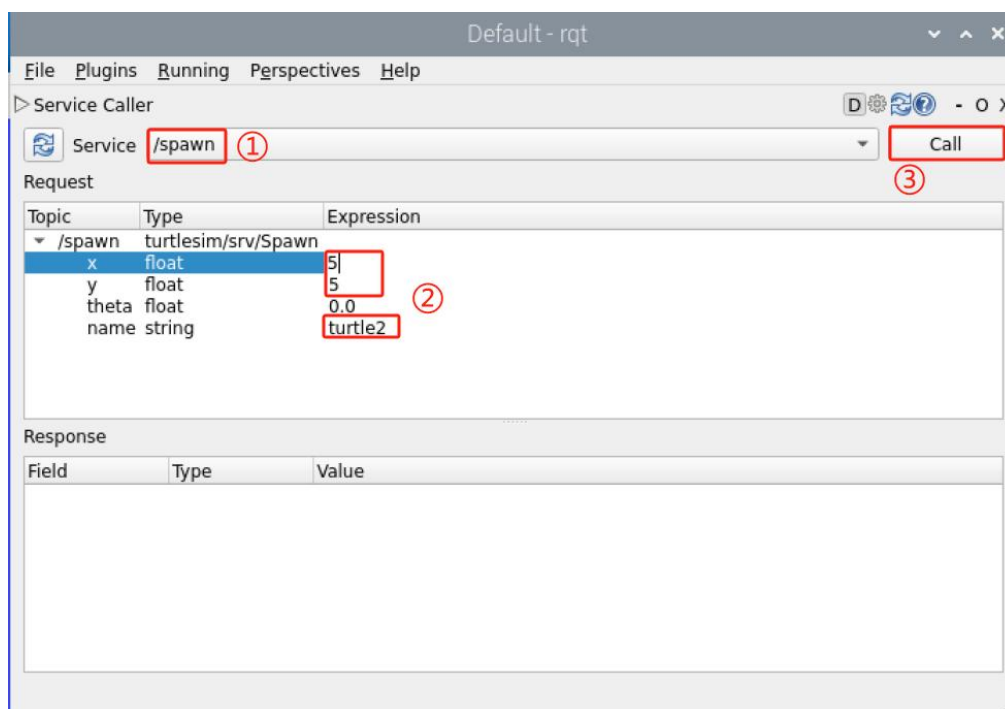


4. Service Plugin

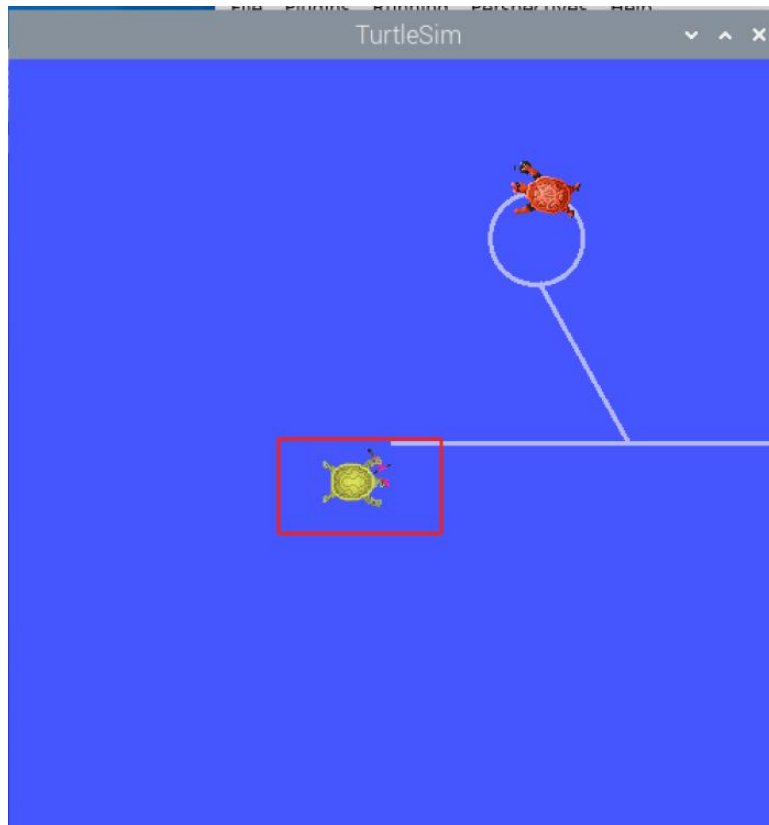
1) Select “**Plugins**→**Services**→**Services Caller**” in sequence to open the service plugin.



2) Select the “**/spawn**” service, then set the position for the new turtle to be spawned, name it as “turtle2”, and finally click on “**Call**”.

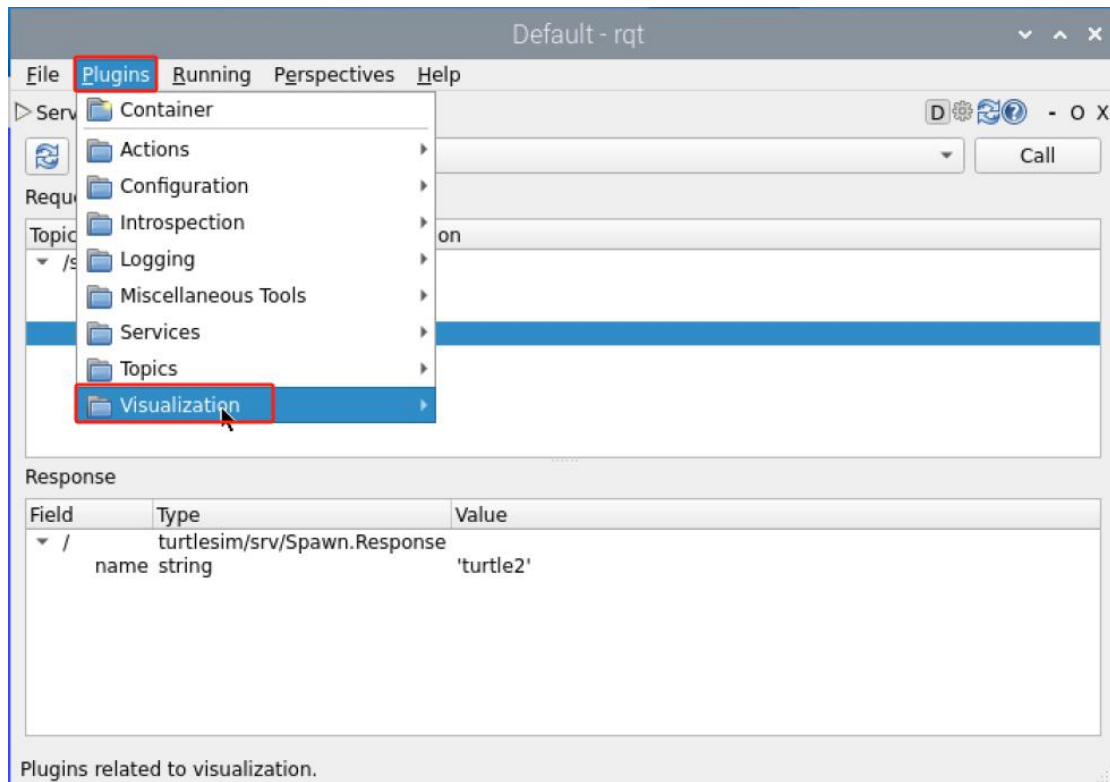


At this point, a turtle named “turtle2” will be spawned.

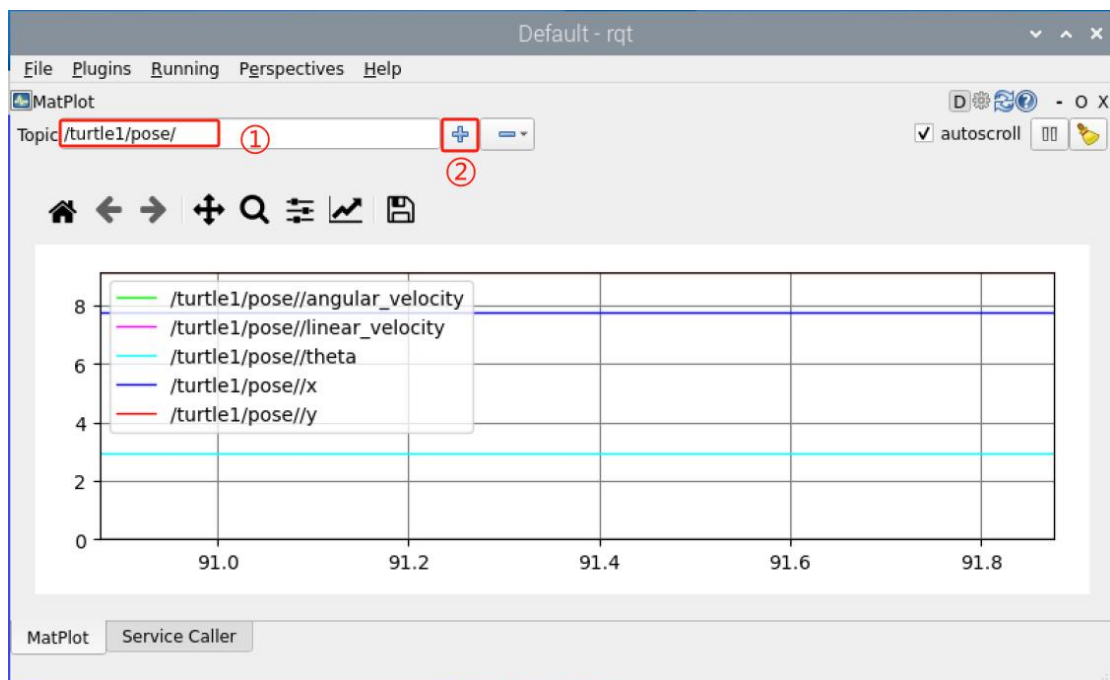


5. Drawing Plugin

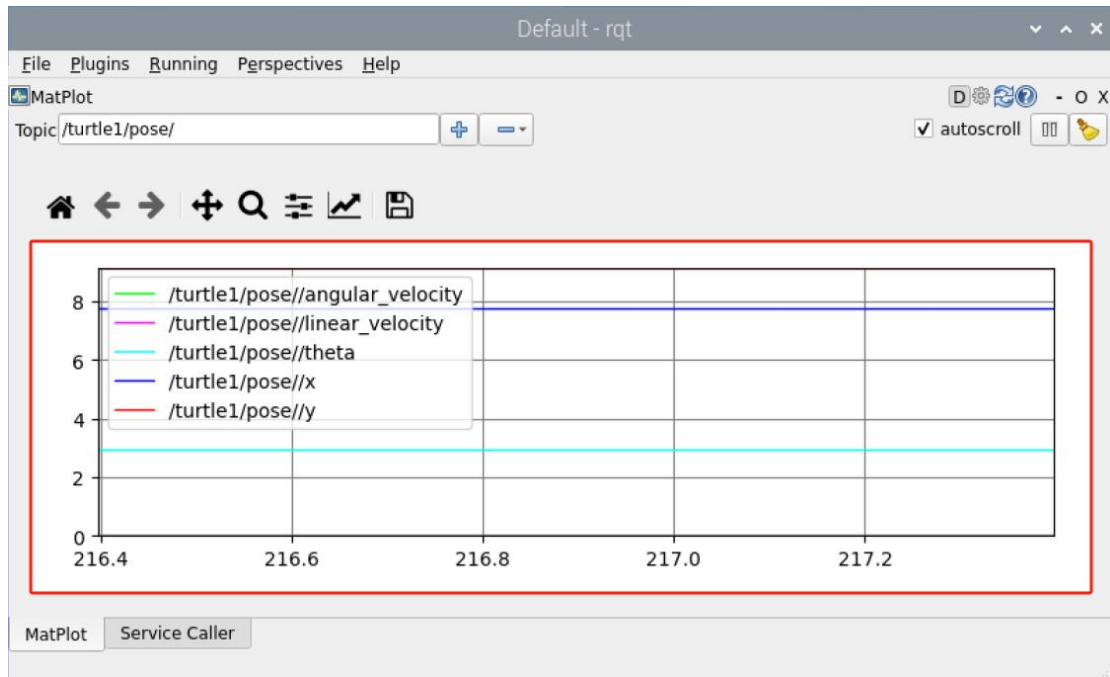
1) Select “Plugins→Visualization→Plot” in sequence to open drawing plugin.



2) Enter `"/turtle/pose/"` to select the turtle's position data, then click .



3) The plugin will visualize the angular velocity, linear velocity, orientation, X coordinate, and Y coordinate of the turtle in real-time.



To further learn about other plugins in the RQT tool, please visit the official website (<https://www.ros.org/>) and explore the tutorials available there.